

MAE6760 Final Project

Prey State Estimation for Dragonfly Hunting

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1 Motivation

Dragonfly pursuit dynamics is a highly complex problem involving unsteady aerodynamics, control theory, and neurobiology. While the wing kinematics leading to prey capture can be observed with high-speed cameras, there remains significant uncertainty as to how the wing muscle commands are formulated based on the available sensory inputs. Some evidence suggests that dragonflies use predictive models of prey behavior to better anticipate prey motion [1]. This would be much more sophisticated than other invertebrate predators like tiger beetles, which appear to use simple proportional control of turn rate based on the angular position of the prey in the field of view [2]. In this study, we introduce a simplified model of dragonfly flight dynamics, and develop a model-based estimation approach for interception of a passive prey with a prescribed trajectory. We compare the results of this approach with a simple proportional control law.

2 Model Description

2.1 Setup and Coordinates

Real insect flight is very much a three-dimensional problem, and involves body rotations about all three axes. The flight dynamics are made very complex by the strong coupling between wing motion and body pitch and roll, which leads to interesting questions about stability, see e.g. [3]. Here, for simplicity, we liken our insects to point masses, and confine their motion to the (x, z) plane, where x is parallel to the ground, and z is normal to the ground, so as to capture the influence of gravity. The resulting motion only has 2 degrees of freedom. In the hunting setup, the prey follows a prescribed motion, and the dragonfly maneuvers to pursue and capture it. In our simple model, the prey does not attempt an escape, in fact it does not react to the dragonfly at all. We go on to describe the dragonfly and prey models.

2.2 Dragonfly Model

We model the dragonfly as a point mass subjected to the force of gravity and to the aerodynamic forces developed by its two pairs of flapping wings. The body motion is 2D, but the wing motion is 3D, and assumed to be symmetrical about the (x, z) plane (left and right wing motion is identical) such that there are no lateral-directional moments and no sideforce. Any pitch moment that would occur in reality due to the inevitable offset between the instantaneous centers of mass and pressure is neglected. This simplification removes an important constraint on the wing kinematics, which is to produce the desired pitch moment (zero over a cycle for steady flight, nonzero for maneuvers). Their only responsibility in our model is to produce the desired body acceleration. With this simplified model, we can write the dragonfly state as $\xi = [x, z, \dot{x}, \dot{z}]^T$.

2.2.1 Aerodynamic Force Model

A major challenge of insect flight modeling is the very complex aerodynamics. Almost the entire range of possible angles of attack may be covered over a single wingbeat, and the reciprocating nature of the motion causes the wings to be strongly affected by their own wakes and those of their neighbors. Here the wings are modeled as flat plates, divided along the span direction into small elements, such that any area distribution can be applied. Lift and drag forces are evaluated at each element. First the relative velocity of the air is projected onto a plane normal to the span direction. Then the angle of attack is evaluated and a simple model for the all-attitude lift and drag coefficients is used:

$$C_D(\alpha) = 1.4 - \cos 2\alpha \quad (1)$$

$$C_L(\alpha) = 1.2 \sin 2\alpha \quad (2)$$

This model, based on Newtonian theory, is due to [4] and was found to be a workable approximation when compared to experiments, even though it completely neglects a plethora of aerodynamic phenomena, including wing-wing interactions. The contributions of each wing element are then summed to obtain the resultant force on the body.

2.2.2 Wing Kinematics

The wing orientation can be described by 4 angles: γ , the stroke plane angle, β , the coning or sweep angle, φ , the stroke or flapping angle, and ψ , the feathering or pitch angle. We will not cover the mathematical formulation in detail. Instead we offer a graphical representation which hopefully makes the definitions of these angles clear. Briefly: γ describes the orientation of the plane traced by the wing tips over a cycle, relative to the horizontal. γ is variable, albeit on a longer time scale than the wingbeat. β is the angle between the span axis and the stroke plane, and is analogous to the sweep angle in aviation. We take it to be constant and equal to 0. Dragonfly forewings usually have a small forward sweep, and hindwings have a small backward sweep, with negligible effects on the aerodynamic force. φ and ψ describe the wing motion within the stroke plane, and are prescribed by sinusoidal functions of time. To limit the number of parameters, we assume that the motion of the forewings and hindwings is identical save for some phase shift σ_0 . The phase shift between

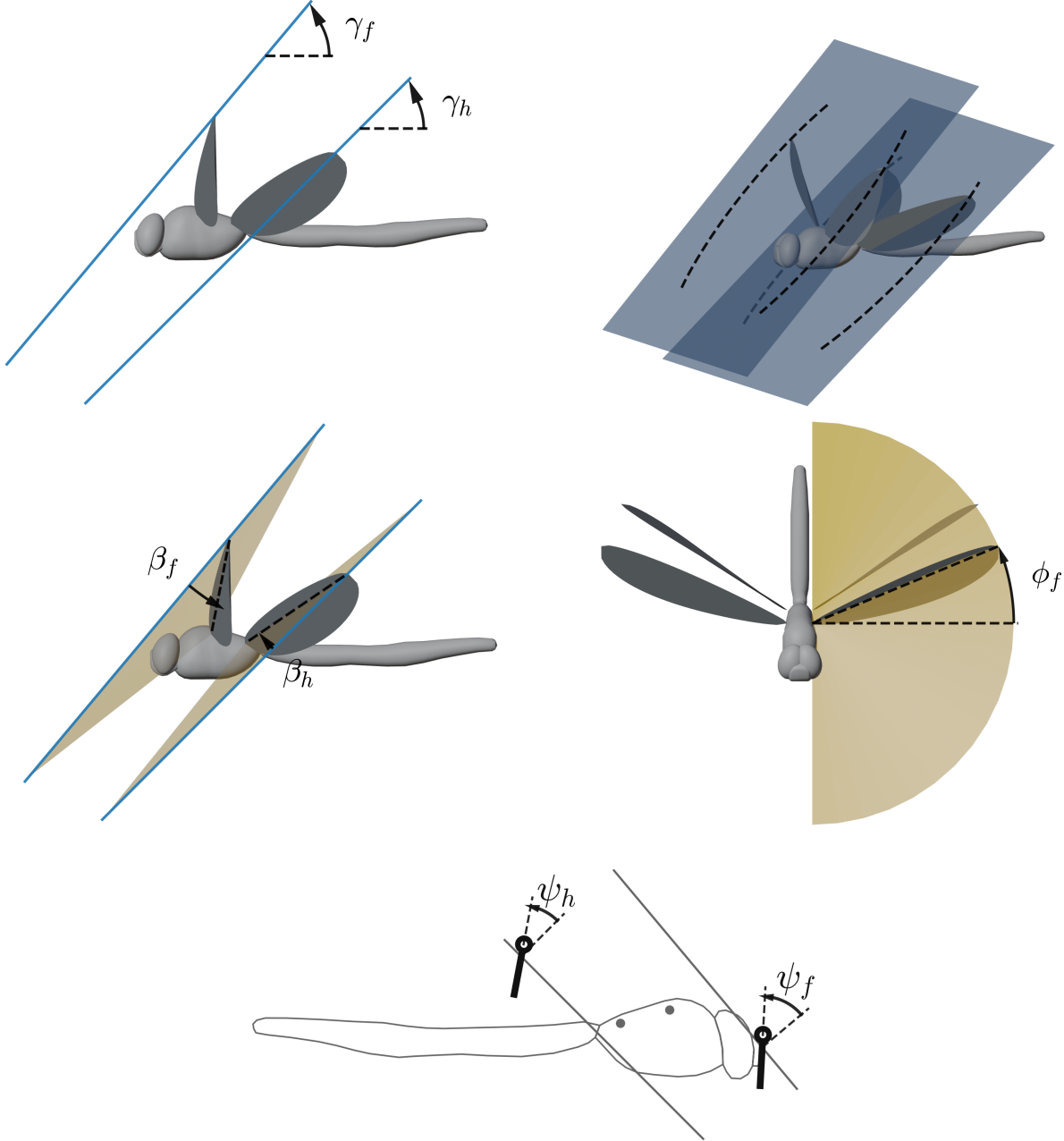
feathering and stroke is δ_0 .

$$\varphi_{fw}(t) = \varphi_1 \cos(\omega_0 t) \quad (3)$$

$$\varphi_{hw}(t) = \varphi_1 \cos(\omega_0 t + \sigma_0) \quad (4)$$

$$\psi_{fw}(t) = \psi_0 + \psi_1 \cos(\omega_0 t + \delta_0) \quad (5)$$

$$\psi_{hw}(t) = \psi_0 + \psi_1 \cos(\omega_0 t + \delta_0 + \sigma_0) \quad (6)$$



2.2.3 Non-dimensionalization

We find it convenient to nondimensionalize the dynamics equation. We choose the dragonfly body length L_b as the length scale, and use gravitational acceleration g to define the time scale as $\mathcal{T} = \sqrt{L_b/g}$. Three nondimensional numbers emerge from this process:

- $\omega_0^* = \omega_0 \sqrt{\frac{L_b}{g}}$: the nondimensional wing beat frequency
- $\mu_0^* = \frac{\rho_{\text{air}} S L_b}{m}$: the nondimensional wing loading, where S is the planform area of one wing, and m is the total mass
- $\lambda_0^* = L_w/L_b$: the nondimensional wing length

In general, the fore- and hindwings may have different values of μ_0^* and λ_0^* , but here we assume they have identical geometries. From literature on dragonfly morphology [5], we find that representative values of the nondimensional parameters are $\omega_0^* = 4\pi$, $\mu_0^* = 0.075$, and $\lambda_0^* = 0.75$. We use these as a baseline.

2.3 Prey Model

We make no attempt to model the aerodynamics of the prey. Instead, we prescribe its motion using a constant-velocity model with white process noise. Taking the prey state to be $\xi_p = [x_p, z_p, \dot{x}_p, \dot{z}_p]^T$, the dynamics are:

$$\ddot{x}_p = w_x \quad (7)$$

$$\ddot{z}_p = w_z \quad (8)$$

where w_x and w_z are independent zero-mean Gaussian processes with (nondimensionalized) intensity q_p^* . With small q_p^* we approach the deterministic constant-velocity case, while large q_p^* makes the trajectory effectively unpredictable beyond a short horizon. We use this same model both to generate the true prey trajectories and as the internal model the dragonfly uses for estimation.

3 Controller Design With and Without State Estimation

3.1 Without State Estimation

For pursuit, given our 2D model, we are primarily concerned with the velocity heading angle θ . The most intuitive control variable for θ is the stroke plane angle γ , since for a symmetric wing stroke (wing pitch angle ψ during the downstroke is the opposite of what it is during the upstroke, i.e. $\psi_0 = 0$), the aerodynamic force direction is normal to the stroke plane. In our design, the wing flapping and pitching amplitudes φ_1 and ψ_1 are fixed to provide high acceleration and a high speed ceiling, and $\dot{\gamma}$ obeys a proportional control law analogous to the tiger beetle turn rate control law from [2]:

$$\dot{\gamma}(t) = -K\theta_e(t - \tau) \quad (9)$$

where K is a proportional gain, $\theta_e = \theta_p - \theta$ in which θ_p is the prey bearing, and τ is the muscle chain delay between the brain issuing a command and the wings actually executing it, which we

take to be half a wingbeat ($\tau/T_{wb} = 0.5$). The compound eye perceptual delay is neglected. Note that while the stroke plane angle is a mode of directional control for dragonflies, it is only secondary compared to body pitch. Since the current model is a simple 2 DoF model, we can consider γ to stand in for both stroke plane tilt and body pitch, since the actual mechanics behind body pitch (complex, due to wing moments) are obscured.

3.2 With State Estimation

In principle, a disadvantage of the proportional control law is that the dragonfly does not anticipate the prey motion, and always steers towards the prey's current bearing. By estimating the prey's state, and converting that estimate into a bearing, the prey capture performance may be improved (by metrics we have yet to define).

3.2.1 Prey State Estimation

We use an extended Kalman filter (EKF) on the prey state ξ_p , propagating it forward under the same constant-velocity model with white-noise acceleration used to generate the trajectory. We assume that the dragonfly is able to keep track of its own state ξ using a variety of sensory organs. The compound eye delivers a noisy prey bearing y_θ to the nervous system

$$y_\theta = \arctan \frac{z_p - z}{x_p - x} + v_\theta, \quad v_\theta \sim \mathcal{N}(0, R_\theta), \quad (10)$$

This is nonlinear in ξ_p , so we linearize about the current mean at each update step. The filter is initialized with a prior $\hat{\xi}_p(0) \sim \mathcal{N}(\bar{\xi}_p, P_0)$. To improve observability, we may want to include information on the apparent size of the prey into the estimate. We will discuss this further in the Results section.

3.2.2 Lead Pursuit

From the estimated prey state and the dragonfly's own state, we compute a lead point. That is, the point where the prey is predicted to be when the dragonfly catches it, assuming the prey continues at its currently-estimated velocity $\hat{v}_p$ and the dragonfly travels at its current speed v in a straight line. The time-to-go t_{go} verifies

$$|\hat{r} + \hat{v}_p t_{go}| = v t_{go}, \quad (11)$$

where $\hat{r} = \hat{p}_p - p$ is the estimated relative position. This is a quadratic in t_{go} with a positive root whenever the dragonfly is fast enough to ever catch the prey. The lead point is then $\hat{p}_p + \hat{v}_p t_{go}$, and the reference bearing θ_ℓ is the angle to that point as seen from the dragonfly. The control law is otherwise identical to simple proportional control

$$\dot{\gamma}(t) = -K\theta_e^{\text{lead}}(t - \tau). \quad (12)$$

Where $\theta_e^{\text{lead}} = \theta_\ell - \theta$. The model-based controller differs from the proportional one only in what bearing it aims for. If there is no valid solution for t_{go} , i.e. the prey moves directly away faster than the dragonfly can chase, it falls back to pure pursuit.

Specific parameter values and additional improvements to the state estimate and lead pursuit strategy will be discussed as results are presented in the following section.

4 Results and Discussion

In comparing the two controllers, we would have liked to provide a more systematic analysis of the parameter dependence of capture rates and other success metrics over a large number of simulations. Because of a number of issues that had to be resolved with the existing dragonfly simulation code, as well as the lead controller code, we present here only a limited and fairly qualitative discussion for two representative cases.

We compared the two controllers for two prey trajectories: a *vertical* case where the prey climbs nearly straight up, and a *horizontal* case where the prey drifts laterally. In both, the dragonfly starts hovering at the origin. For the vertical case the prey appears at $(6L_b, 0)$ with nominal velocity $(0, 1)$. For the horizontal case the prey appears at $(3L_b, 3L_b)$ with nominal velocity $(1, 0)$. The process noise in both cases is $q_p^* = 0.1$. Both controllers see the same prey realization within each case. The wing stroke kinematic parameters other than the stroke plane angle γ are constant and selected from past studies to provide good acceleration and a high speed ceiling. γ is initialized at 30° , a value representative of dragonfly hover. The control law gain is set to $K = 1.5$, a value considered optimal based on previous studies of the proportional controller. The muscle delay is set to $\tau/T_{wb} = 0.5$. All these parameters are common to both the proportional and MBE controllers. The MBE filter takes angular-size measurements with $R_\theta = 10^{-2}$, and a noisy prior $P_0 = 0.2^2 I_4$. Capture is declared at separation $\leq 0.5L_b$.

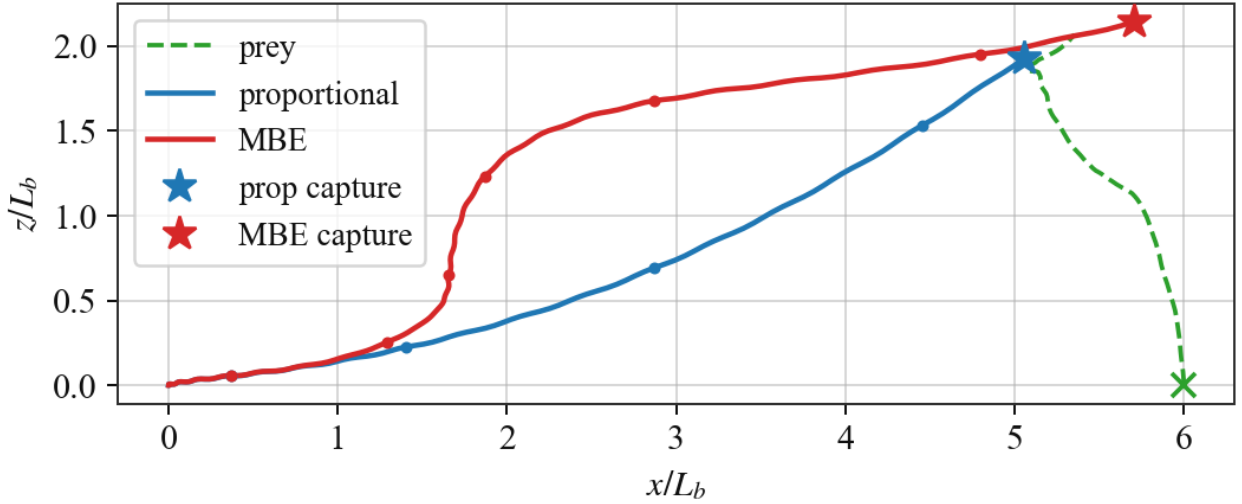


Figure 2: Vertical prey trajectory

It is quite obvious from the vertical case trajectories and time traces that the MBE lead pursuit controller does not perform as well as expected. Both controllers capture: the proportional controller at $t/T_{wb} \approx 17.5$, the MBE controller at 25.7, both at the capture threshold of $0.5L_b$. The proportional path is quite smooth. The MBE path starts off slightly steeper as expected, but suddenly decides to pitch up severely. It appears that the sudden pitch up command coincides (with delay) with a jump in the vertical velocity error. The dragonfly thinks the prey is a lot faster than it is, and way overcorrects to lead its position. Eventually the error settles a little, but remains quite significant overall. This aside, the estimator appears to perform reasonably well in terms of consistency.

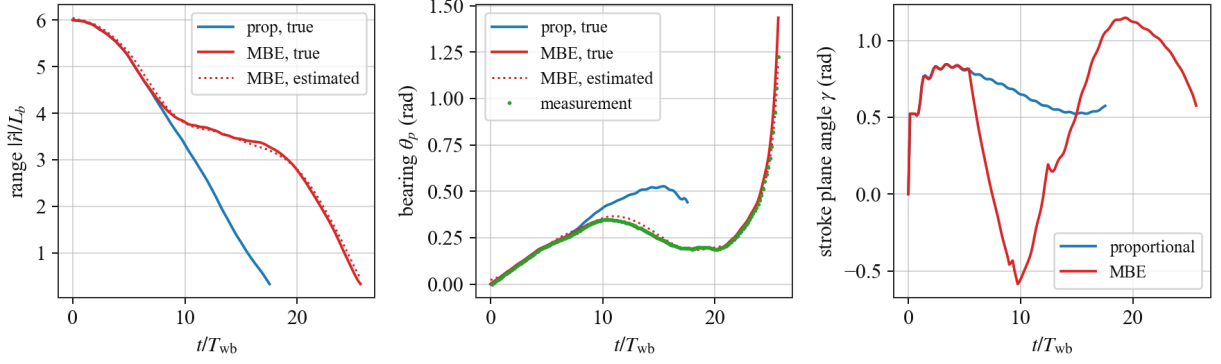


Figure 3: Vertical case time series

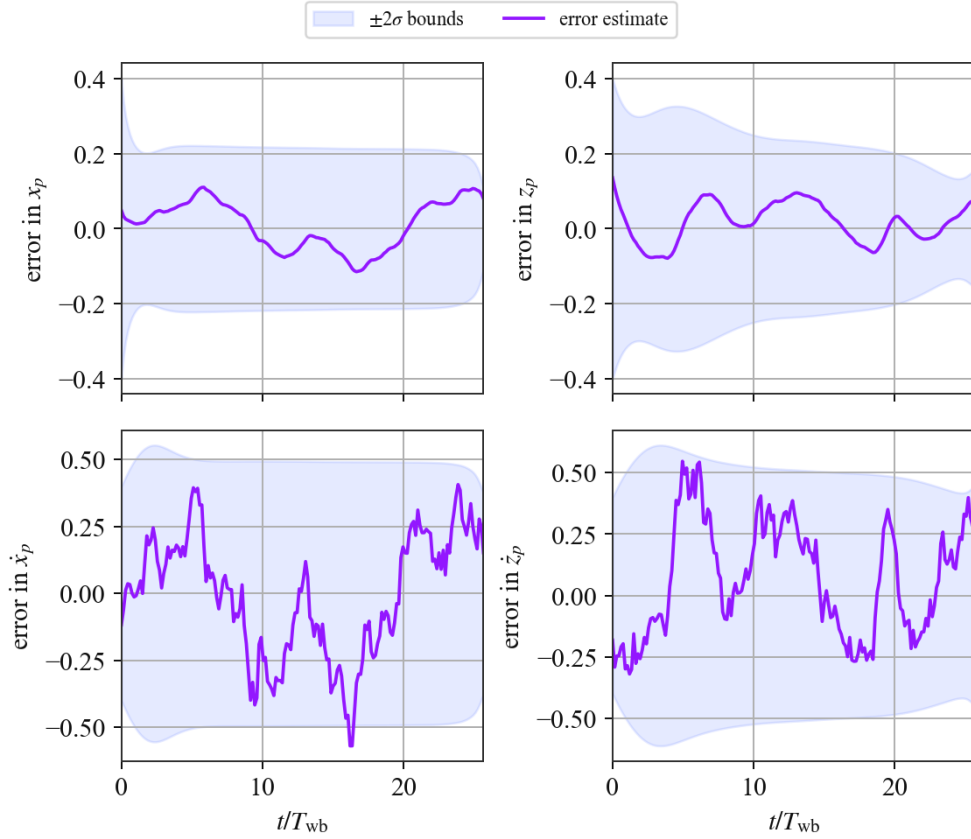


Figure 4: Vertical case estimator error

The MBE controller is quite a bit more successful in the horizontal case, and notably manages an intercept, whereas the proportional controller is unsuccessful. In light of these two cases, it appears the MBE controller has potential but is quite fragile. Even the dynamics of this highly simplified dragonfly flight model are surprisingly rich, and since the dragonfly's own state is obviously an integral part of the prey state estimation, the dynamics can interact with the controller in unpredictable and consequential ways.

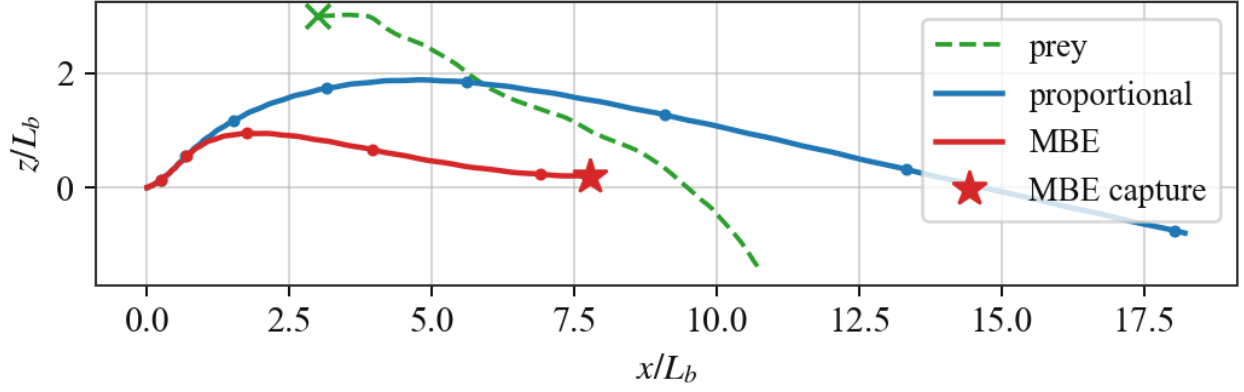


Figure 5: Horizontal prey trajectory

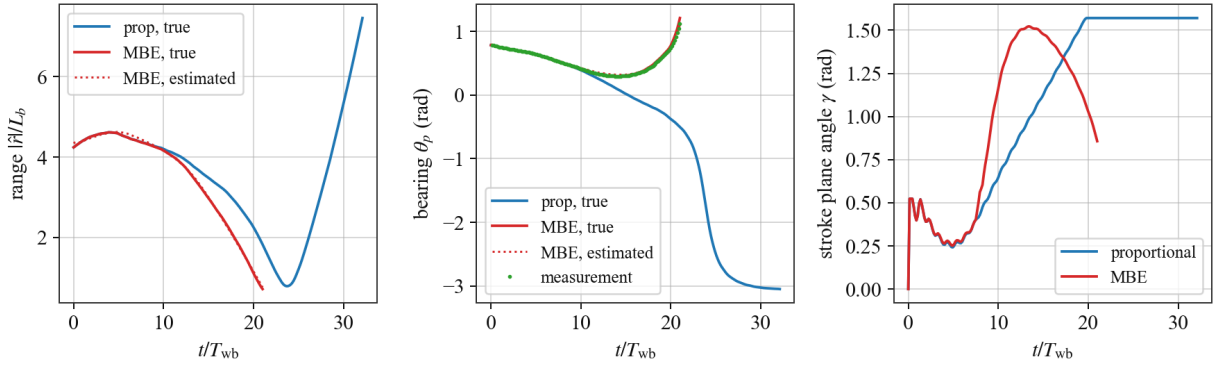


Figure 6: Horizontal case time series

Since all this is hard to reason about from first principles, an essential next step would be to conduct systematic sweeps of the parameter space, first for given morphology and kinematics, then over the morphological and kinematic parameters themselves, for a large number of simulated prey trajectories, to get a better understanding of the controller dynamics. Since two drivers of bad performance appear to be large estimator error and time delay, it would also make sense to think about improving the estimate by fusing range or range rate measurements (from angular size of prey), and somehow compensating for actuation lag within the controller.

References

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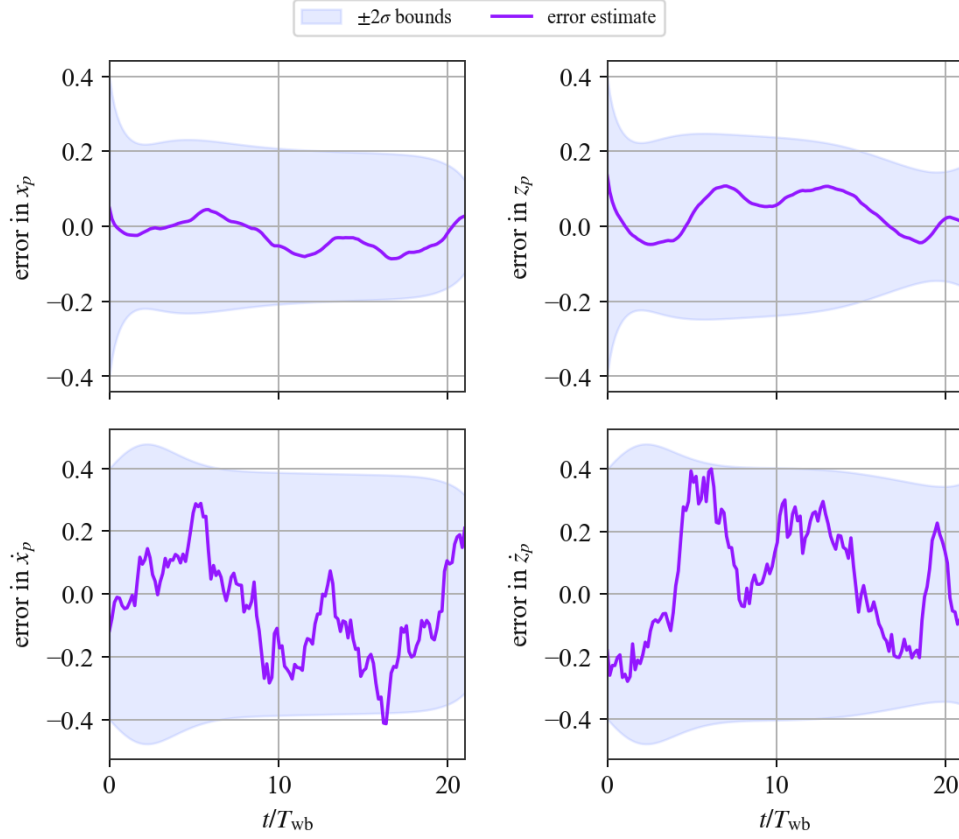


Figure 7: Horizontal case estimator error

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